

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (IT)

April/May 2012

IT 202 Data Communication

Date: 02.05.2012, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) The noise figure of an amplifier is 7dB. Calculate the output signal to noise ratio when the input signal to noise ratio is 35 dB. [02]
- (b) Low frequency noise is also called as a _____. [01]
- (c) The tuned circuit of the oscillator in a simple AM transmitter employs a 50-microhenry coil and a 1-nanofarad capacitor. If the oscillator output is modulated by audio frequencies up to 10kHz, what is the frequency range occupied by the sidebands? [03]
- (d) Define Amplitude Modulation. [01]
- Q - 2 (a) The antenna current of an AM broadcast transmitter, modulated to a depth of 40 percent by an audio sine wave, is 11A. it increase to 12A as a result of simultaneous modulation by another audio sine wave. What is the modulation index due to this second wave? [06]
- (b) A 400 – watt (400W) carrier is modulated to a depth of 75 percent. Calculate the total power in the modulated wave. [03]
- (c) Write short note on noise temperature. [05]

OR

- Q - 2 (a) A certain transmitter radiates 9kW with the carrier unmodulated, and 10.125kW when the carrier is sinusoidally modulated. Calculate the modulation index, percent of modulation. If another sine wave, corresponding to 40 percent modulation, is transmitted simultaneously, determine the total radiated power. [06]
- (b) A broad cast radio transmitter radiates 10 kilowatts when the modulation percentage is 60. How much of this is carrier power? [03]
- (c) Write short note on partition noise. [05]
- Q - 3 (a) What is tracking in receiver? Give its different type and explain it with mathematical equation. [06]
- (b) An AM broad cast receiver has an IF of 465 kHz and is tuned to 1000kHz, and the RF stage has one tuned circuit with a Q of 50. [04]
- (A) Find the image frequency.
- (B) Find the image rejection in decibels.
- (c) Derive the friis's formula for N number of amplifier connected in cascade connection. [04]

OR

- Q - 3 (a) Explain the block diagram of double-conversion superhetrodyne receiver and draw its frequency response at different stage. [06]

- (b) A receiver tunes signals from 550 to 1600kHz with an IF of 455 kHz. Find the frequency tuning ranges and capacitor tuning ranges for the oscillator section and for the RF section. [04]
- (c) Write short note on Johnson noise. [04]

SECTION – II

- Q - 4 (a) Explain mutual inductance with suitable equation of series and parallel coil. [05]
- (b) List the types of stability in oscillator and explain amplitude stability. [02]
- Q - 5 (a) Write short note on voltage controlled oscillator. [06]
- (b) Calculate the noise factor of an attenuator pad that has that has an insertion loss of 6dB. [02]
- (c) Write short note on high frequency transformers. [06]

OR

- Q - 5 (a) Define attenuator pad and explain T-attenuator & pi-attenuator. [06]
- (b) Draw the basic block diagram of phase locked loop frequency synthesizer and explain it. [06]
- (c) Define noise figure. [02]
- Q - 6 (a) Draw the hybrid model for three different configuration of bipolar junction transistor. [08]
- (b) Write short note on different properties of Fourier transform. [06]

OR

- Q - 6 (a) What is periodic and aperiodic signal? Compare power and energy signals. [06]
- (b) Draw the generalized hybrid model for BJT and derive following parameter. [08]
1. current gain
 2. input impedance
 3. voltage gain
 4. output impedance
